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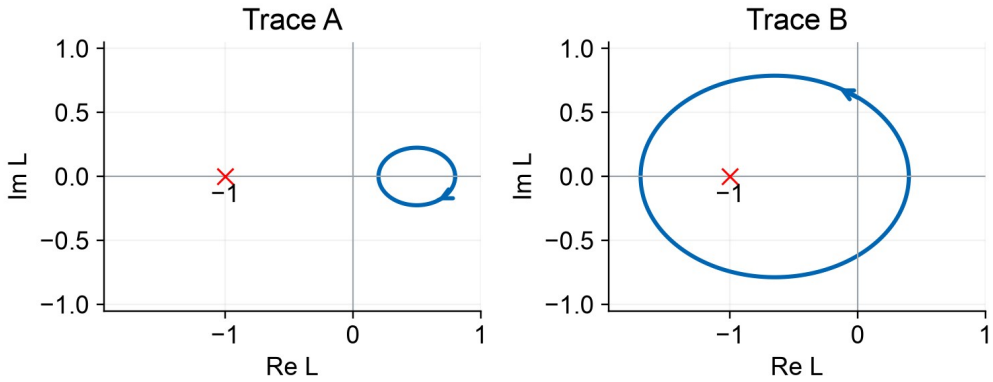
Tutorial 08

Nyquist decisions and robustness evidence

Preparation: Lecture 8: closed-loop stability, complete Nyquist contours, open-loop unstable poles, margins and their limitations.
Submission credit: 1% of the course grade for submitted work; correctness is not marked.
In class: Q1 and Q2 cover Nyquist decisions, margins and one given-delay calculation. Extension A is an optional teacher demonstration of the cubic proof; Extension B contains optional delay-budget and gain-change calculations.

Question 1 A Nyquist trace needs an open-loop pole count

The supplied plots are schematic COMPLETE Nyquist traces, including the negative-frequency branch; their arrows give traversal direction. Neither trace passes through −1, and there are no imaginary-axis open-loop poles. Use clockwise encirclements as positive: N is clockwise count minus counterclockwise count, and Z = P + N. P counts open-loop right-half-plane poles; Z counts closed-loop right-half-plane poles.



a) For records (A, P = 0), (A, P = 1), (B, P = 1) and (B, P = 2), calculate N and Z and state stability.

b) A fifth record claims P = 0 and N = −1. Explain why this cannot be a valid complete-contour record under the stated convention. What would you recheck?

Question 2 How much delay can a camera-alignment loop tolerate?

A nominal stable loop has no open-loop right-half-plane poles and has single relevant crossovers. Its gain crossover is 6 rad/s with phase −138°. At its phase crossover of 15 rad/s, magnitude is −12 dB. A proposed pure sensing delay is 0.025 s. Require remaining phase margin of at least 25°. Treat delay and gain changes separately unless a full frequency-response curve is supplied.

a) Find the nominal phase margin in degrees and the gain margin in dB and as a gain factor.

b) Find the phase-margin loss from the proposed delay and the remaining phase margin. Does it meet the 25° requirement?

MATLAB verification. Plot remaining phase margin against added delay using $PM(\tau) = PM_0 - \omega_{gc}\tau \times 180/\pi$. Mark the proposed 0.025 s delay and the 25° requirement. Report phase loss and remaining margin. Base MATLAB is sufficient. Maximum delay is optional Extension B.

Extension A A gain family with no finite negative-axis crossing

Optional teacher demonstration / strengthening only. The third-order proof is not required for the core tutorial.

A separate normalized unity-feedback loop is given below. You may argue from how the polynomial roots move continuously as K varies; a Routh table is not required.

$$L(s)=\frac{K(s+2)}{(s+1)(s+3)(s+5)},\ K>0$$

a) Separate real and imaginary parts of L(jω) and determine whether its positive-frequency trace crosses the negative real axis at a finite nonzero frequency.

b) Derive the closed-loop cubic. Test for a possible imaginary-axis root as K varies and decide stability for all positive finite K.

c) Does the resulting gain-margin observation prove satisfactory phase margin, transient response or robustness?

Extension B Delay budget and gain change

Optional. Use the same crossover records and single-crossover assumptions as Q2.
a) Calculate maximum total delay and the additional allowable delay beyond 0.025 s for a remaining phase margin of 25°.
b) In the no-added-delay loop, multiply the gain by 1.2. Calculate the new gain margin and decide whether the new phase margin can be obtained from only the two original records.

Method reference

Method sources: Ogata, Modern Control Engineering, 5th ed., Problem A-7-7, pp. 528–529 (Nyquist reasoning). New contour records, margin data and gain family are used. Page numbers refer to printed pages.
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